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Open Geospatial Consortium

NWS CONNECT SERVICE PROFILE

DRAFT STANDARD
Implementation

APPROVED

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I ABSTRACT

This document defines the NWS Connect Service Profile, an OGC API — Environmental Data Retrieval (EDR) Part 3 Service Profile. It specifies normative requirements and conformance tests for server implementations conforming to this profile.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC API, Features, NWS Connect, water gauge, real-time, pub/sub, service profile



PREFACE

This document was prepared by NOAA/NWS/MDL.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this document.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- NOAA/NWS/MDL

VI SUBMITTERS

All questions regarding this document should be directed to the editor or the submitters:

Table — Submitters

NAME	AFFILIATION
Shane Mill *(editor)*	NOAA/NWS/MDL

VII POINT OF CONTACT

This service is provided by NOAA/NWS/Meteorological Development Laboratory (MDL) (https://www.weather.gov/mdl/). Email: nws.mdl.support@noaa.gov
Address: USA
Contact instructions: Data provided by the National Weather Service.

1 SCOPE

1 SCOPE

This standard defines the NWS Connect Service Profile. It specifies requirements and conformance tests for implementations of this profile.

2 CONFORMANCE

2 CONFORMANCE

Conformance with this standard shall be checked using the Abstract Test Suite in Annex A.

The following conformance classes are defined:

- http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/nws_connect

3

NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Mark Burgoyne, David Blodgett, Charles Heazel, Chris Little, Open Geospatial Consortium: OGC 19-086r6, *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium (2023). http://www.opengis.net/doc/IS/ogcapi-edr-1/1.1.0.

OGC API — EDR Part 3: Service Profiles (draft)

4 TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

This document also uses terms defined in OGC API — EDR Part 1: Core.

5 REQUIREMENTS

5 REQUIREMENTS

NOTE: Conformance with this profile is assessed against its requirements ("shall") only. Recommendations ("should") and permissions ("may") are provided as informative guidance: they are not conformance-tested and an implementation is not made non-conformant by not following them. See the quality-assessment summary emitted by the profile tooling for a count of the recommendations and permissions defined by this profile.

REQUIREMENTS CLASS 1	
IDENTIFIER	`http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
CONFORMANCE CLASS	Conformance class A.1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/nws_connect`
TARGET-TYPE	NWS Connect Service Profile Profile Standard
NORMATIVE STATEMENTS	Requirement 1: `/req/nws_connect/conformance` Requirement 2: `/req/nws_connect/collections-endpoint` Requirement 3: `/req/nws_connect/water-gauge-collection` Requirement 4: `/req/nws_connect/wwa-collection` Requirement 5: `/req/nws_connect/space-weather-collection` Requirement 6: `/req/nws_connect/cwa-collection` Requirement 7: `/req/nws_connect/states-collection` Requirement 8: `/req/nws_connect/items-by-id` Requirement 9: `/req/nws_connect/pubsub-amqp` Requirement 10: `/req/nws_connect/pubsub-websocket` Requirement 11: `/req/nws_connect/pubsub-filters-water-gauge` Requirement 12: `/req/nws_connect/pubsub-filters-wwa` Requirement 13: `/req/nws_connect/pubsub-message-attributes-wwa` Requirement 14: `/req/nws_connect/pubsub-filters-space-weather`

REQUIREMENT 1	
IDENTIFIER	`/req/nws_connect/conformance`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The service SHALL declare conformance to OGC API — EDR Part 1 Core and Part 2 PubSub.
A	The service SHALL provide a /conformance endpoint.

REQUIREMENT 1	
B	The response SHALL include http://www.opengis.net/spec/ogcapi-edr-1/1.0/conf/core.
C	The response SHALL include http://www.opengis.net/spec/ogcapi-edr-2/1.0/conf/pubsub.

REQUIREMENT 2	
IDENTIFIER	`/req/nws_connect/collections-endpoint`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The service SHALL provide a /collections endpoint listing all collections.
A	The service SHALL provide a /collections endpoint.
B	The response SHALL include water_gauge, wwa, space_weather, cwa, and states collections.
C	Each collection SHALL have id, title, description, extent, and links properties.
D	The /collections/{collectionId}/queryables endpoint SHALL expose Part 3 custom dimensions (e.g. for wwa, alert_action / ibw_tier / max_hail_size_in / max_wind_gust_mph / parent_item_id) so partners can discover filterable attributes.

REQUIREMENT 3	
IDENTIFIER	`/req/nws_connect/water-gauge-collection`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The water_gauge collection SHALL provide GeoJSON features with flood stage information.
A	The service SHALL provide a /collections/water_gauge endpoint.
B	The service SHALL provide a /collections/water_gauge/items endpoint.
C	The response Content-Type SHALL be application/geo+json.
D	Each feature SHALL include flood_stage, feet_above_stage, and value properties.
E	Each feature geometry SHALL be a GeoJSON Point.

REQUIREMENT 4

IDENTIFIER	`/req/nws_connect/wwa-collection`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The wwa collection SHALL provide GeoJSON features for weather watches and warnings.
A	The service SHALL provide a /collections/wwa endpoint.
B	The service SHALL provide a /collections/wwa/items endpoint.
C	The response Content-Type SHALL be application/geo+json.
D	Each feature SHALL include CAP_info_event, CAP_info_severity, CAP_info_urgency, and CAP_status properties.
E	Each feature geometry SHALL be a GeoJSON Polygon or MultiPolygon.
F	Each SVR / TOR / SVS feature SHALL include alert_action ('initial', 'upgrade', 'update', 'cancel', or 'other').
G	Each SVR / TOR / SVS feature SHALL include ibw_tier reflecting the current Impact-Based Warning tier per NWSI 10-511.
H	Each SVR / TOR / SVS feature SHALL include parent_item_id resolving to the initial SVR or TOR that started the lifecycle.
I	Each SVR / TOR / SVS feature SHALL include vtec_action, vtec_etn, vtec_office, vtec_phenomenon, and vtec_significance parsed from the P-VTEC line(s).
J	Each SVR / TOR / SVS feature SHALL include max_hail_size_in and max_wind_gust_mph (0 when unknown) so numeric filter policies apply uniformly.
K	Each SVR / TOR / SVS feature MAY include a banner object with text, bg (background color), fg (foreground color), and severity (1-5) when a life-safety banner applies (TORNADO EMERGENCY, PARTICULARLY DANGEROUS SITUATION, DESTRUCTIVE / CONSIDERABLE DAMAGE THREAT, TORNADO POSSIBLE, TORNADO OBSERVED, FLASH FLOOD EMERGENCY).
L	Each feature MAY include a states property listing two-letter USPS state codes covered by the warning geometry, derived from CAP UGC and SAME geocodes.

REQUIREMENT 5

IDENTIFIER	`/req/nws_connect/space-weather-collection`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`

REQUIREMENT 5	
STATEMENT	The space_weather collection SHALL provide GeoJSON features for NOAA SWPC space weather alerts.
A	The service SHALL provide a /collections/space_weather endpoint.
B	The service SHALL provide a /collections/space_weather/items endpoint.
C	The response Content-Type SHALL be application/geo+json.
D	Each feature SHALL include message_type, noaa_scale, scale_category, and scale_level properties.

REQUIREMENT 6	
IDENTIFIER	`/req/nws_connect/cwa-collection`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The cwa collection SHALL provide GeoJSON features for NWS County Warning Areas.
A	The service SHALL provide a /collections/cwa endpoint.
B	The service SHALL provide a /collections/cwa/items endpoint.
C	The response Content-Type SHALL be application/geo+json.
D	Each feature SHALL include a CWA identifier property.
E	Each feature geometry SHALL be a GeoJSON Polygon or MultiPolygon.

REQUIREMENT 7	
IDENTIFIER	`/req/nws_connect/states-collection`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The states collection SHALL provide GeoJSON features for US state boundaries.
A	The service SHALL provide a /collections/states endpoint.

REQUIREMENT 7	
B	The service SHALL provide a /collections/states/items endpoint.
C	The response Content-Type SHALL be application/geo+json.
D	Each feature SHALL include a state identifier property.
E	Each feature geometry SHALL be a GeoJSON Polygon or MultiPolygon.

REQUIREMENT 8	
IDENTIFIER	`/req/nws_connect/items-by-id`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	All collections SHALL support retrieval of individual features by ID.
A	Each collection SHALL provide a /collections/{collectionId}/items/{featureId} endpoint.
B	The response SHALL return a single GeoJSON feature.
C	The response SHALL return 404 if the feature does not exist.

REQUIREMENT 9	
IDENTIFIER	`/req/nws_connect/pubsub-amqp`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The service SHALL support AMQP 1.0 and MQTT pub/sub for real-time notifications on water_gauge, wwa, and space_weather collections.
A	The service SHALL provide an AMQP 1.0 broker (edr-broker) on port 5672 (RabbitMQ 4.x native AMQP 1.0).
B	The service SHALL provide an MQTT broker (edr-broker) on port 1883 over the same amq.topic exchange.
C	Clients SHALL be able to consume from a queue bound to amq.topic with binding key collections.water_gauge.items.
D	Clients SHALL be able to consume from a queue bound to amq.topic with binding key collections.wwa.items.

REQUIREMENT 9	
E	Clients SHALL be able to consume from a queue bound to amq.topic with binding key collections. space_weather.items.
F	Clients SHALL authenticate with SASL PLAIN using their account email as the username and a named WebSocket key (wsk_...) as the password.
G	Notifications SHALL be published as GeoJSON features.

REQUIREMENT 10	
IDENTIFIER	`/req/nws_connect/pubsub-websocket`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The service SHALL support a native WebSocket pub/sub endpoint for browser clients.
A	The service SHALL provide a WebSocket endpoint at /ws for local development (ws) and production (wss).
B	The WebSocket endpoint SHALL deliver the same GeoJSON feature notifications as the AMQP 1.0 and MQTT transports.
C	WebSocket connections SHALL authenticate with the same identity used for the AMQP/MQTT transports.

REQUIREMENT 11	
IDENTIFIER	`/req/nws_connect/pubsub-filters-water-gauge`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The water_gauge collection pub/sub SHALL support filtering by flood stages, stations, and WFO.
A	Subscriptions SHALL accept flood_stages filter (array of stage values).
B	Subscriptions SHALL accept stations filter (comma-separated station IDs or 'all').
C	Subscriptions SHALL accept wfo filter (Weather Forecast Office codes).
D	Subscriptions SHALL accept email filter for notification delivery.

REQUIREMENT 12	
IDENTIFIER	`/req/nws_connect/pubsub-filters-wwa`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The wwa collection pub/sub SHALL support filtering by event types, state, CWA, delivery method, warning lifecycle, and Impact-Based Warning thresholds.
A	Subscriptions SHALL accept event_types filter (array of weather event types).
B	Subscriptions SHALL accept state filter (comma-separated USPS state codes or blank for all).
C	Subscriptions SHALL accept cwa filter (comma-separated County Warning Area codes).
D	Subscriptions SHALL accept email filter for notification delivery.
E	Subscriptions SHALL accept a delivery_method value of either 'email' (default) or 'sms'.
F	Subscriptions with delivery_method='sms' SHALL require a phone_number filter (any common US format; the service normalizes to E.164).
G	Subscriptions with delivery_method='sms' SHALL require at least one of state or cwa to be set (to bound message volume and per-message cost).
H	Subscriptions SHALL accept alert_action filter with value 'initial' (default), 'upgrade', or 'both' selecting which lifecycle events fire a notification.
I	Subscriptions SHALL accept ibw_tier_min filter (CONSIDERABLE, DESTRUCTIVE, PDS, EMERGENCY) to floor the Impact-Based Warning tier that triggers alerts.
J	Subscriptions SHALL accept hail_min_in and wind_min_mph numeric filters; when both are set they combine via logical OR so either threshold fires the alert.
K	When a subscription opts in to upgrade re-alerts on a Severe Thunderstorm Warning or Tornado Warning event type, the service SHALL automatically include 'Severe Weather Statement' in the effective event filter so follow-up SVS products can trigger the alert.

REQUIREMENT 13	
IDENTIFIER	`/req/nws_connect/pubsub-message-attributes-wwa`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	Each wwa pub/sub notification SHALL carry a set of message attributes that enable partner-side and broker-side filtering across all supported transports (SNS filter policies, AMQP 1.0 message properties, MQTT subscription topics).

REQUIREMENT 13	
A	Each notification SHALL carry event, severity, urgency, certainty, wfo, and sender String attributes.
B	Each notification SHALL carry alert_action, ibw_tier, ibw_prior_tier, vtec_phenomenon, vtec_significance, vtec_action, vtec_etn, and parent_item_id String attributes when applicable.
C	Each notification SHALL carry max_hail_size_in and max_wind_gust_mph Number attributes (0 when unknown) so numeric filter policies can express threshold-based subscriptions.
D	Each notification MAY carry a banner_text String attribute when a life-safety banner applies.
E	Each notification SHALL carry a states String.Array attribute listing the two-letter USPS codes covered by the warning geometry.
F	Each notification SHALL carry a dedup_token String attribute keyed by (parent_item_id, ibw_tier, alert_action) so downstream paging systems cannot collapse a genuine upgrade re-alert as a duplicate of the parent warning.

REQUIREMENT 14	
IDENTIFIER	`/req/nws_connect/pubsub-filters-space-weather`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
STATEMENT	The space_weather collection pub/sub SHALL support filtering by message type and NOAA scale.
A	Subscriptions SHALL accept message_type filter (Alert, Warning, Watch, Summary).
B	Subscriptions SHALL accept scale_category filter (G, S, R).
C	Subscriptions SHALL accept scale_level filter (minimum NOAA scale level 1-5).
D	Subscriptions SHALL accept email filter for notification delivery.

A

ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

A ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

CONFORMANCE CLASS A.1	
IDENTIFIER	`http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/nws_connect`
REQUIREMENTS CLASS	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nws_connect`
CONFORMANCE TESTS	Abstract test A.1-1: `/conf/nws_connect/conformance` Abstract test A.1-2: `/conf/nws_connect/collections-endpoint` Abstract test A.1-3: `/conf/nws_connect/water-gauge-collection` Abstract test A.1-4: `/conf/nws_connect/wwa-collection` Abstract test A.1-5: `/conf/nws_connect/space-weather-collection` Abstract test A.1-6: `/conf/nws_connect/cwa-collection` Abstract test A.1-7: `/conf/nws_connect/states-collection` Abstract test A.1-8: `/conf/nws_connect/items-by-id` Abstract test A.1-9: `/conf/nws_connect/pubsub-amqp` Abstract test A.1-10: `/conf/nws_connect/pubsub-websocket` Abstract test A.1-11: `/conf/nws_connect/pubsub-filters-water-gauge` Abstract test A.1-12: `/conf/nws_connect/pubsub-filters-wwa` Abstract test A.1-13: `/conf/nws_connect/pubsub-message-attributes-wwa` Abstract test A.1-14: `/conf/nws_connect/pubsub-filters-space-weather`

Example 1

label	/conf/nws_connect/conformance
subject	/req/nws_connect/conformance
test-purpose	Validate that conformance is correctly implemented.

Send GET request to /conformance.

Verify response contains http://www.opengis.net/spec/ogcapi-edr-1/1.0/conf/core.

Verify response contains http://www.opengis.net/spec/ogcapi-edr-2/1.0/conf/pubsub.

Example 2

label	/conf/nws_connect/collections-endpoint
subject	/req/nws_connect/collections-endpoint
test-purpose	Validate that collections endpoint is correctly implemented.

Send GET request to /collections.

Verify response contains collections array.

Verify water_gauge, wwa, space_weather, cwa, and states are present in the collections array.

Verify each collection has id, title, description, extent, and links.

Example 3

label	/conf/nws_connect/water-gauge-collection
subject	/req/nws_connect/water-gauge-collection
test-purpose	Validate that water gauge collection is correctly implemented.

Send GET request to /collections/water_gauge.

Verify response contains collection metadata.

Send GET request to /collections/water_gauge/items?limit=10.

Verify response Content-Type is application/geo+json.

Verify each feature contains flood_stage, feet_above_stage, and value properties.

Verify each feature geometry type is Point.

Example 4

label	/conf/nws_connect/wwa-collection
subject	/req/nws_connect/wwa-collection
test-purpose	Validate that wwa collection is correctly implemented.

Send GET request to /collections/wwa.

Verify response contains collection metadata.

Send GET request to /collections/wwa/items?limit=10.

Verify response Content-Type is application/geo+json.

Verify each feature contains CAP_info_event, CAP_info_severity, CAP_info_urgency, and CAP_status properties.

Verify each feature geometry type is Polygon or MultiPolygon.

Example 5

label	/conf/nws_connect/space-weather-collection
subject	/req/nws_connect/space-weather-collection
test-purpose	Validate that space weather collection is correctly implemented.

Send GET request to /collections/space_weather.

Verify response contains collection metadata.

Send GET request to /collections/space_weather/items?limit=10.

Verify response Content-Type is application/geo+json.

Verify each feature contains message_type, noaa_scale, scale_category, and scale_level properties.

Example 6

label	/conf/nws_connect/cwa-collection
subject	/req/nws_connect/cwa-collection
test-purpose	Validate that cwa collection is correctly implemented.

Send GET request to /collections/cwa.

Verify response contains collection metadata.

Send GET request to /collections/cwa/items?limit=10.

Verify response Content-Type is application/geo+json.

Verify each feature contains a CWA identifier.

Verify each feature geometry type is Polygon or MultiPolygon.

Example 7

label	/conf/nws_connect/states-collection
subject	/req/nws_connect/states-collection
test-purpose	Validate that states collection is correctly implemented.

Send GET request to /collections/states.

Verify response contains collection metadata.

Send GET request to /collections/states/items?limit=10.

Verify response Content-Type is application/geo+json.

Verify each feature contains a state identifier.

Verify each feature geometry type is Polygon or MultiPolygon.

Example 8

label	/conf/nws_connect/items-by-id
subject	/req/nws_connect/items-by-id
test-purpose	Validate that items by id is correctly implemented.

Send GET request to /collections/water_gauge/items?limit=1.

Extract the id of the first feature.

Send GET request to /collections/water_gauge/items/{featureId}.

Verify response is a single GeoJSON feature with matching id.

Send GET request to /collections/water_gauge/items/nonexistent.

Verify response status is 404.

Example 9

label	/conf/nws_connect/pubsub-amqp
subject	/req/nws_connect/pubsub-amqp
test-purpose	Validate that pubsub amqp is correctly implemented.

Connect to the AMQP 1.0 broker (edr-broker) on port 5672 using SASL PLAIN (email + wsk_key).

Declare a queue and bind it to the amq.topic exchange with binding key collections.water_gauge.items.

Verify notifications are received as GeoJSON features.

Bind a queue to amq.topic with binding key collections.wwa.items.

Verify notifications are received as GeoJSON features.

Connect to the MQTT broker (edr-broker) on port 1883 and subscribe to topic collections/+/items.

Verify notifications are received as GeoJSON features.

Example 10

label	/conf/nws_connect/pubsub-websocket
subject	/req/nws_connect/pubsub-websocket
test-purpose	Validate that pubsub websocket is correctly implemented.

Connect to the native WebSocket endpoint at /ws (ws locally, wss in production).

Verify the connection succeeds and delivers GeoJSON feature notifications.

Verify notifications match those delivered over the AMQP 1.0 and MQTT transports.

Example 11

label	/conf/nws_connect/pubsub-filters-water-gauge
subject	/req/nws_connect/pubsub-filters-water-gauge
test-purpose	Validate that pubsub filters water gauge is correctly implemented.

Subscribe to collections/water_gauge/items/# with flood_stages filter set to [“major”, “moderate”].

Verify only notifications matching the specified flood stages are received.

Subscribe with stations filter set to specific station IDs.

Verify only notifications for those stations are received.

Subscribe with wfo filter set to a specific office code.

Verify only notifications for that WFO are received.

Example 12

label	/conf/nws_connect/pubsub-filters-wwa
subject	/req/nws_connect/pubsub-filters-wwa
test-purpose	Validate that pubsub filters wwa is correctly implemented.

Subscribe to collections/wwa/items/# with event_types filter set to [“Tornado Warning”].

Verify only notifications matching the specified event types are received.

Subscribe with state filter set to a specific state code.

Verify only notifications for that state are received.

Subscribe with cwa filter set to a specific CWA code.

Verify only notifications for that CWA are received.

Subscribe with delivery_method=’sms’ and no state or cwa filter; verify the service rejects the subscription with a 4xx error identifying the missing scope.

Subscribe with delivery_method=’sms’, phone_number=’(555) 234-5678’, state=’TN’; verify the subscription is accepted and the phone number is normalized to ‘+15552345678’ in the persisted record.

Subscribe with alert_action=’both’, event_types=[“Severe Thunderstorm Warning”]; verify the service transparently adds “Severe Weather Statement” to the effective event filter so follow-up SVS products can fire the alert.

Publish a synthetic SVR followed by an SVS that raises the IBW tier from BASE to CONSIDERABLE; verify a subscription with alert_action=’upgrade’ receives exactly one notification for the SVS and none for a subsequent CON that keeps the tier at CONSIDERABLE.

Subscribe with hail_min_in=1.75 and wind_min_mph=70; verify notifications for a product with maxHailSize=2.5 (hail matches) and separately for a product with maxWindGust=75 (wind matches) both arrive (OR semantics).

Verify notifications for a product with maxHailSize=1.0 and maxWindGust=60 do NOT arrive.

Example 13

label	/conf/nws_connect/pubsub-message-attributes-wwa
subject	/req/nws_connect/pubsub-message-attributes-wwa
test-purpose	Validate that pubsub message attributes wwa is correctly implemented.

Subscribe to collections/wwa/items and receive a live SVR notification.

Verify the notification carries event, severity, urgency, certainty, wfo, sender String message attributes.

Verify the notification carries alert_action, ibw_tier, vtec_action, vtec_etn, parent_item_id String message attributes.

Verify the notification carries max_hail_size_in and max_wind_gust_mph Number message attributes (present even when zero).

Verify the notification carries a states String.Array message attribute matching the state(s) covered by the warning geometry.

Verify the notification carries a dedup_token String message attribute formatted as '<parent_item_id>|<ibw_tier>|<alert_action>'.

For an SVR followed by an SVS upgrade, verify the two notifications have DIFFERENT dedup_token values so paging systems cannot collapse them as duplicates.

Example 14

label	/conf/nws_connect/pubsub-filters-space-weather
subject	/req/nws_connect/pubsub-filters-space-weather
test-purpose	Validate that pubsub filters space weather is correctly implemented.

Subscribe to collections/space_weather/items/# with message_type filter set to "Warning".

Verify only notifications matching the specified message type are received.

Subscribe with scale_category filter set to "G".

Verify only geomagnetic-storm notifications are received.

Subscribe with scale_level filter set to 3.

Verify only notifications at or above NOAA scale level 3 are received.